

Smart Waste Segregation System

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Introduction

Smart Waste Segregation Systems use deep learning, computer vision, and sensors to automatically identify and separate waste into categories such as plastic, paper, metal, glass, and organic waste. Automated segregation improves recycling efficiency, reduces human effort, and supports sustainable waste management.

Objectives

- Improve recycling accuracy.
- Reduce landfill waste.
- Minimise manual sorting.
- Promote smart city initiatives.

Working Principle

The system captures an image of waste using a camera. A trained deep learning model, such as a Convolutional Neural Network (CNN), classifies the waste type. Based on the prediction, a controller activates the appropriate bin or sorting mechanism. The system can also record statistics for monitoring recycling performance.

Key Components

Hardware: Camera, conveyor belt, servo motors, microcontroller (Arduino/Raspberry Pi), sensors, smart bins.

Software: Python, TensorFlow/PyTorch, OpenCV, NumPy, Flask or Streamlit dashboard.

Advantages

- High classification accuracy.
- Faster waste processing.
- Reduced operational costs.

- Environmentally friendly and scalable.

Applications

Smart waste segregation is used in municipal waste management, hospitals, educational institutions, shopping malls, industries, airports, and smart cities. It helps increase recycling rates while reducing environmental pollution and greenhouse gas emissions.

Future Scope

Future systems can integrate IoT, robotic arms, cloud analytics, and AI-powered monitoring to improve accuracy and provide real-time waste statistics. Renewable energy-powered smart bins and predictive analytics can further enhance sustainable waste management.

Conclusion

A Smart Waste Segregation System is an intelligent solution that combines artificial intelligence and automation to improve waste handling. By ensuring accurate waste classification and efficient recycling, it contributes to cleaner cities, resource conservation, and a more sustainable future.